

David Osorio

832-487-6377 | osoriodav2@gmail.com | osoriodav2.github.io

EDUCATION

Wesleyan University

Bachelors of Arts

Double Major in Computer Science, Mathematics, GPA 3.71/4.0

Relevant Coursework: Discrete Mathematics, Applied Data Analysis, Computer Science I/II

Middletown, CT

May 2028

Houston City College

Associate of Science

Major in Computer Science, GPA: 3.97/4.0; Highest Honors Graduate; Honors College Graduate

Relevant Coursework: Data Structures and Algorithms, Computer Organization and Architecture

Houston, TX

May 2025

RESEARCH EXPERIENCE

NSF Undergraduate Researcher, University of Houston-Downtown, Houston, TX

June 2026 – Present

- Developed a Python script using pandas and scikit-learn to analyze Differential Privacy (ϵ) vs. model utility.
- Integrated UCI machine learning datasets to analyze both LDP and CDP models.
- Benchmarked private regressions against baseline models to model accuracy retention.

Student Researcher, Wesleyan University, Middletown, CT

Jan 2026

- Conducted quantitative research on a national U.S. survey (N=929) to analyze demographic and regional predictors of climate change beliefs.
- Executed bivariate and multivariate analyses (Chi-square, ANOVA, logistic/linear regression) using R.

PROJECT EXPERIENCE

NYC Crime Explorer

Apr 2026

- Developed an interactive web map analyzing NYPD criminal complaints and U.S. Census ACS data, color-coded by offense severity.
- Built dynamic frontend filters and a backend Python script (pandas, scipy) to run regression analyses against demographic controls.

Stock Prediction Agent

Mar 2026

- Designed an automated quantitative trading agent in Python using the Yahoo Finance API to model and forecast equity price movements.
- Built a robust backtesting framework to simulate historical strategies and evaluate model performance.
- Engineered a real-time module to generate trading insights and optimize risk based on historical volatility.

WORK EXPERIENCE

Service Desk Student Worker, Wesleyan University, Middletown, CT

Sept 2025 – Present

- Resolved technical and access issues for faculty and alumni regarding Duo, Outlook, and Gmail.
- Diagnosed hardware and software problems for students to restore productivity across campus systems and printers.

IT Technician, University of Houston at Katy, Houston, TX

Oct 2024 – Aug 2025

- Troubleshoot lecture hall hardware, software, and audio-visual systems for faculty and students.
- Maintained critical campus infrastructure, including library computing networks and printing systems.

HONORS AND AWARDS

Citation of Citizenship in the Department of Mathematics and Computer Science

May 2026

ASA DataFest CT 2026 Category Winner

Apr 2026

Ronald E. McNair Scholar

Nov 2025

Phi Theta Kappa Membership

Feb 2025

Houston City College Honors College

July 2024

AP Scholar with Distinction

July 2024

Texas First Early High School Graduation Scholar

May 2024

ADDITIONAL

Technical Skills: R, RStudio, C, C++, Python (pandas, scikit-learn, matplotlib, numpy), Windows Server, Network Management, Access Control Lists

Certifications & Training: CompTIA Network+ CE, Rice University Environmental Data Academy

Languages: Fluent in English, Spanish